

Discrete Drainage Dynamics VII: Gravity–Gauge Coupling and the Joint Wilson-Loop Back-Reaction

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May 2026

Abstract

Paper II derives gravity from the back-reaction of scalar drainage on the substrate, with mobility $\mu(R)$. Paper VI derives a candidate fixed-point formula for α_{EM} from the back-reaction of a Wilson loop along the body diagonal of the unit cube. Both mechanisms share the same physical substrate; the present paper examines what their coupling implies. We argue that on the unit cube, two distinct back-reaction loops coexist—the body-diagonal Wilson loop (length $\sqrt{3}$) carrying α_{EM} , and the cube-surface Wilson surface (area 6) carrying the gravitational coupling at the substrate scale. Under the assumption that the two sectors share the same combinatorial prefactor and one-loop structure, we obtain a leading *tree-level structural* ratio $\alpha_G^{\text{lat}}/\alpha_{\text{EM}}^{\text{lat}} = \sqrt{3/6} = 1/\sqrt{2}$. Anchoring this ratio with the laboratory low-energy value $\alpha_{\text{EM}}(0) = 1/137.036$ yields a phenomenological substrate mass scale $m_{\text{lat}} \approx 0.072 M_P$ and lattice spacing $\ell_{\text{lat}} \approx 14 \ell_P$ (a full running-coupling treatment is deferred). Under a minimal local-reserve coupling ansatz $\varepsilon_{\text{local}}(R) = \varepsilon_0 R/R_0$, two cross-coupling effects follow: a gravitational modulation of α_{EM} proportional to $|\Phi_{\text{grav}}|/c^2$ near massive bodies, and an EM-induced drainage proportional to the local Maxwell stress-energy. Cosmological drift of α_{EM} inherited from a slowly-varying R_0 is potentially in tension with current atomic-clock bounds for an $O(1)$ Hubble fraction, providing the most stringent immediate falsifier. This construction reframes the apparent weakness of gravity as a *scale hierarchy*: at the substrate scale the two couplings are geometrically comparable, while ordinary particles lie far below that scale. The paper proposes a structural closure between Papers II and VI: one substrate, one back-reaction principle, two emergent couplings sharing a common geometric origin.

1 Main claims and scope

1. **One substrate, two back-reaction loops.** The gravitational and electromagnetic sectors of DDD share the same cubic substrate. On the unit cube, two distinct closed Wilson objects coexist: the body diagonal (a 1D loop of squared length 3) carrying the U(1) gauge phase of Paper VI, and the cube surface (a 2D Wilson surface of area 6) carrying the gravitational drainage of Paper II. They are *not* the same loop, and they back-react on different geometric quantities.
2. **Tree-level structural ratio at the substrate scale.** Under the assumption that the two sectors share the same combinatorial prefactor $8\pi^2 = 4 \text{Vol}(S^3)$ and the same one-loop structure but with different characteristic lengths, the leading structural ratio is

$$\boxed{\frac{\alpha_G^{\text{lat}}}{\alpha_{\text{EM}}^{\text{lat}}} = \sqrt{\frac{x_{\text{EM}}}{x_{\text{grav}}}} = \sqrt{\frac{3}{6}} = \frac{1}{\sqrt{2}}.} \quad (1)$$

This is a leading structural prediction conditional on the shared Wilson-loop prefactor; it is not yet verified at one loop in the joint sector (§4).

3. **Phenomenological substrate mass scale at $\sim 0.072 M_P$.** Combining (1) with the laboratory low-energy value $\alpha_{\text{EM}}(0) = 1/137.036$ (used here as a phenomenological anchor; a full running-coupling treatment between the substrate scale and laboratory scale is deferred) and the defining relation $\alpha_G(m) = (m/M_P)^2$ gives

$$\frac{m_{\text{lat}}}{M_P} = \sqrt{\frac{1}{137.036 \sqrt{2}}} \approx 0.072, \quad (2)$$

which fixes the substrate's characteristic mass at $\sim 0.072 M_P \approx 8.8 \times 10^{17} \text{ GeV}$ and the lattice length scale at $\ell_{\text{lat}} \approx 14 \ell_P$.

4. **Two cross-coupling consequences from a minimal ansatz.** Under the minimal local-reserve coupling ansatz $\varepsilon_{\text{local}}(R) = \varepsilon_0 R/R_0$, the shared substrate generates two leading-order cross-coupling effects:

(i) *Gravity modulates α_{EM} :* the body-diagonal Wilson loop near a mass M sees a depleted reserve $R(r)/R_0 = \sqrt{1 - \beta/r}$ (Paper II), shifting the effective x_{EM}^* and hence α_{EM} by an amount of order $\delta\alpha/\alpha \sim |\Phi_{\text{grav}}|/c^2$ at distance r from the source.

(ii) *EM modulates G :* the local Maxwell stress-energy contributes to substrate drainage just as ordinary mass-energy does, generating a small additional gravitational potential proportional to $\rho_{\text{EM}} = (E^2 + B^2)/(8\pi)$.

Limitations and scope. We do not derive the absolute value of Newton's G in SI units; that is the calibration result of Paper II. We do not compute Standard Model fermion masses; their hierarchy relative to m_{lat} is the subject of Paper XV. We do not address how strong or weak nuclear interactions emerge. The cross-coupling magnitudes derived below are leading-order estimates based on the back-reaction structure inherited from Papers II and VI; precision predictions require analytical closure of the open problems listed in those papers (the small-world correction ε and the exact one-loop coefficient V_{deff}).

Motivation and significance

One of the long-standing puzzles of fundamental physics is the enormous apparent weakness of gravity compared with electromagnetism at ordinary particle scales. For two electrons, the gravitational coupling is smaller than the electromagnetic one by roughly forty orders of magnitude. In standard units this hierarchy is usually expressed through the small ratio m_e/M_P , but the question remains: why should gravity be so weak for the particles we observe?

The present paper proposes a DDD interpretation of this hierarchy. Gravity and electromagnetism are not treated as unrelated mechanisms. They are two back-reaction channels of the same discrete substrate, associated with two distinct closed objects on the unit cube: a body-diagonal Wilson loop for the gauge sector and a cube-surface Wilson surface for scalar drainage. At the substrate scale, their leading structural ratio is not exponentially small but geometric,

$$\frac{\alpha_G^{\text{lat}}}{\alpha_{\text{EM}}^{\text{lat}}} = \frac{1}{\sqrt{2}},$$

conditional on the shared Wilson-loop prefactor.

In this reading, gravity is not intrinsically weak at the substrate scale. It appears weak at laboratory scales because ordinary particle masses lie far below the substrate mass scale. Anchoring the structural ratio with the low-energy value of α_{EM} gives a phenomenological substrate

scale $m_{\text{lat}} \sim 0.07 M_P$, and the smallness of gravitational effects for electrons is shifted into the mass hierarchy m_e/m_{lat} .

This paper therefore does not claim a complete unification of forces. It proposes a more limited *structural bridge*: one substrate, one back-reaction principle, and two emergent couplings associated with different closed geometric objects. The remaining tasks are explicit: compute the joint one-loop correction, derive the local reserve-dependence of the effective dimension, and (in Paper XV) explain the hierarchy between Standard Model masses and m_{lat} .

The same bridge produces falsifiable consequences. If the local reserve modifies the effective small-world dimension, then massive bodies should induce a small modulation of α_{EM} , and cosmological evolution of the reference reserve R_0 should induce a secular drift of α_{EM} . The hierarchy puzzle, the gravity–gauge relation, and possible variation of fundamental constants thus become linked to the same substrate coupling and become testable on a common experimental front. Figure 1 summarises the scale picture.

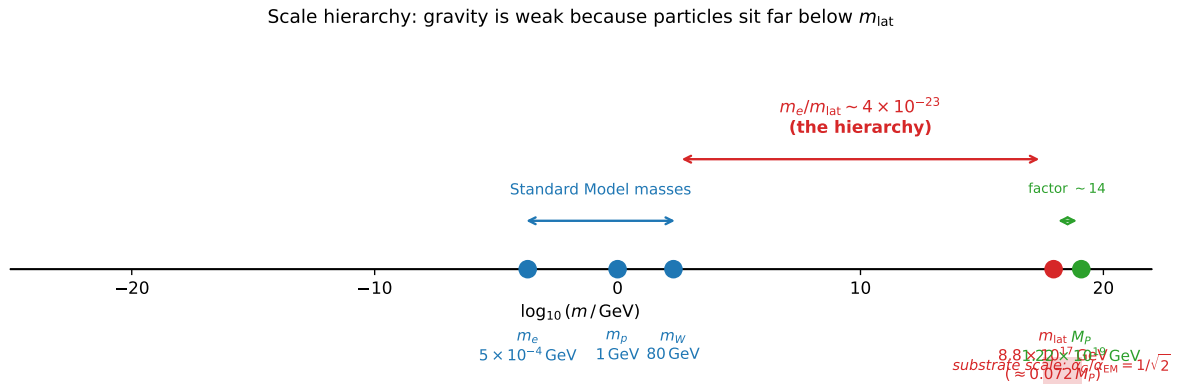


Figure 1: Scale hierarchy interpretation. Standard Model masses sit at $\sim 10^{-3}$ to 10^2 GeV, ~ 23 orders of magnitude below the substrate mass scale $m_{\text{lat}} \approx 0.072 M_P \approx 8.8 \times 10^{17}$ GeV. The substrate is itself a factor ~ 14 below the Planck mass. At the substrate scale, gravity and electromagnetism are comparable ($\alpha_G/\alpha_{\text{EM}} = 1/\sqrt{2}$); the observed weakness of gravity for ordinary particles is a consequence of the ratio $m_e/m_{\text{lat}} \sim 4 \times 10^{-23}$, not of any fundamental weakness of the gravitational coupling.

2 Physical picture in plain words

The mystery. Gravity looks absurdly weak compared with electromagnetism at ordinary scales. Two electrons repel electrically far more strongly than they attract gravitationally — by roughly forty orders of magnitude. In DDD, Paper VII suggests that this weakness may not be fundamental. At the substrate scale, gravity and electromagnetism may be two nearly comparable back-reactions of the same discrete medium. The enormous hierarchy appears only when one asks how a very light particle, such as the electron, samples that substrate.

Imagine the substrate of the previous papers as a vast 3-dimensional jungle gym of nodes and links. Paper II said: at each node lives a reserve of energy that can drain locally, and when a mass concentrates somewhere, the drainage rate is modified, which back-reacts on the local clock-rate. That gives gravity. Paper VI said: each link also carries a phase, and when you trace the smallest closed body-diagonal loop, the resulting Wilson loop self-energy fixes the strength of the electromagnetic coupling. That gives α_{EM} .

The natural question. If both stories live on the same substrate, do they interact? The answer, in DDD, is yes, and the interaction is geometric, not added by hand.

Two loops on one cube. The body diagonal of the unit cube has squared length 3. The

boundary of the cube (its six faces) has total area 6. These are the two simplest closed objects you can build out of one cube of substrate. The body-diagonal loop carries the gauge sector and gives α_{EM} . The cube-surface object carries the drainage flux through the cube’s faces and gives the gravitational coupling at the substrate scale, α_G^{lat} . Same prefactor (it comes from the same combinatorial multiplication $4 \times \text{Vol}(S^3) = 8\pi^2$), but a different characteristic length squared. So:

$$\alpha_{\text{EM}}^{\text{lat}} = \frac{1}{8\pi^2\sqrt{3}}, \quad \alpha_G^{\text{lat}} = \frac{1}{8\pi^2\sqrt{6}}, \quad \frac{\alpha_G^{\text{lat}}}{\alpha_{\text{EM}}^{\text{lat}}} = \frac{1}{\sqrt{2}}.$$

What scale is this? A coupling of order 5×10^{-3} for gravity is enormous compared to its everyday value (electron-to-electron gravity has $\alpha_G \sim 10^{-45}$). But the everyday value is evaluated for particles whose masses are tiny compared with the substrate scale. At the substrate, gravity and electromagnetism are within a factor $\sqrt{2}$ of each other. *The hierarchy is therefore not placed in the relative strength of the two substrate couplings, but in the ratio between observed particle masses and m_{lat} .* Anchoring against the laboratory $\alpha_{\text{EM}}(0)$ places the substrate at $m_{\text{lat}} \approx 0.07 M_P$, about 14 Planck lengths in linear extent (subject to a future running-coupling refinement).

Cross-coupling. Because the two back-reactions live on the same substrate, each can modulate the other—once we fix how the local reserve R enters the small-world correction ε . Under the simplest linear ansatz, a nearby mass depletes R , which shifts the body-diagonal loop’s effective length, which shifts α_{EM} . An electromagnetic field carries energy, which contributes to drainage, which shifts the gravitational potential. Both effects are calculable in principle; both are below current sensitivity at laboratory potentials but the cosmological version is already in the window of atomic-clock bounds.

3 Setup: the two sectors on a shared substrate

3.1 The gravitational sector (recap of Paper II)

Each node i carries a non-negative scalar reserve $R_i \geq 0$. The local update rule is gradient-descent of a drainage functional with mobility $\mu(R)$; in the presence of a mass-energy distribution $\rho(\mathbf{x})$, the steady-state solution gives the radial profile

$$R^2(r)/R_0^2 = 1 - \frac{\beta}{r}, \quad \beta = 2GM/c^2, \quad (3)$$

which is the Schwarzschild g_{00} component of GR (Paper II, §5–7).

3.2 The gauge sector (recap of Paper VI)

Each node carries an additional orientation phase $\alpha_i \in [0, 2\pi)$, each link carries a U(1) gauge phase $\psi_{ij} \in [0, 2\pi)$, and the local energy

$$E_{\text{orient}} = \sum_{(i,j)} J_{ij} [1 - \cos(\alpha_j - \alpha_i + \psi_{ij})] \quad (4)$$

gives compact U(1) lattice gauge dynamics. The body-diagonal Wilson loop self-energy yields

$$\alpha_{\text{EM}}^* = \frac{1}{8\pi^2\sqrt{x_*}}, \quad x_* = 3 + \alpha_{\text{EM}}\sqrt{3}(1 - \alpha_{\text{EM}}V_{d_{\text{eff}}}), \quad (5)$$

matching CODATA at sub-ppm level for $d_{\text{eff}} = 3 + \varepsilon$ with $\varepsilon \in \{1/(3\pi), 1/(2\pi), 1/6\}$ (Paper VI, §5).

3.3 The shared substrate

The same nodes carry both R_i and α_i ; the same links carry both the link-state of Paper III and the gauge phase ψ_{ij} . The combined energy is

$$E_{\text{tot}} = E_{\text{reserve}}[R] + E_{\text{orient}}[\alpha, \psi], \quad (6)$$

and the dynamics is gradient descent of (6) with the appropriate mobility for each sector. Cross-coupling arises because the local substrate state R_i enters μ_{ij} , the mobility of the link, which can in principle depend on the neighbouring orientation phases as well. We do not specify the exact form of $\mu(R, \alpha, \psi)$ here; we examine the leading consequences of the simplest natural coupling.

4 Joint Wilson-loop back-reaction

4.1 Two distinct closed objects on the unit cube

The unit cube admits two natural closed Wilson objects:

Body-diagonal loop. The body diagonal connects opposite corners through the cube’s interior, with squared Euclidean length 3. There are four such diagonals per cube. This is the elementary Wilson *loop* of the gauge sector. Closure along this loop provides the perimeter for the α_{EM} self-consistency.

Cube surface. The boundary of the cube comprises six square faces, total area 6. There are four cubes sharing any given node. The cube surface is the elementary Wilson *surface* of the scalar sector: drainage flux out of the cube must close. Surface area 6 provides the “perimeter” for the gravitational self-consistency.

These two objects are geometrically distinct but share the same substrate combinatorics, namely the four body-diagonal lines and the spinor S^3 phase space (Paper III). Figure 2 illustrates the two elementary closed Wilson objects. The shared combinatorial prefactor is

$$N_{\text{geom}} = 4 \times \text{Vol}(S^3) = 4 \times 2\pi^2 = 8\pi^2, \quad (7)$$

identical for both sectors *by hypothesis*; this assumption is part of what §4–5 are testing.

4.2 Coupling structures from the two loops

For each loop type, we postulate the same back-reaction form (5). At lowest order (tree level, before including the back-reaction):

$$\alpha_{\text{EM}}^{(0)} = \frac{1}{N_{\text{geom}} \sqrt{x_{\text{EM}}}} = \frac{1}{8\pi^2 \sqrt{3}}, \quad (8)$$

$$\alpha_G^{\text{lat},(0)} = \frac{1}{N_{\text{geom}} \sqrt{x_{\text{grav}}}} = \frac{1}{8\pi^2 \sqrt{6}}. \quad (9)$$

The structural ratio is therefore

$$\frac{\alpha_G^{\text{lat},(0)}}{\alpha_{\text{EM}}^{(0)}} = \sqrt{\frac{3}{6}} = \frac{1}{\sqrt{2}}, \quad (10)$$

fixed by geometry alone, conditional on the shared prefactor. We refer to this as the *tree-level structural ratio* of the joint sector. Whether this ratio survives at one loop is conditional on the loop coefficients of the two sectors matching, which is not yet established.

Two distinct closed objects on the unit cube share the same prefactor $8\pi^2$ but different characteristic lengths

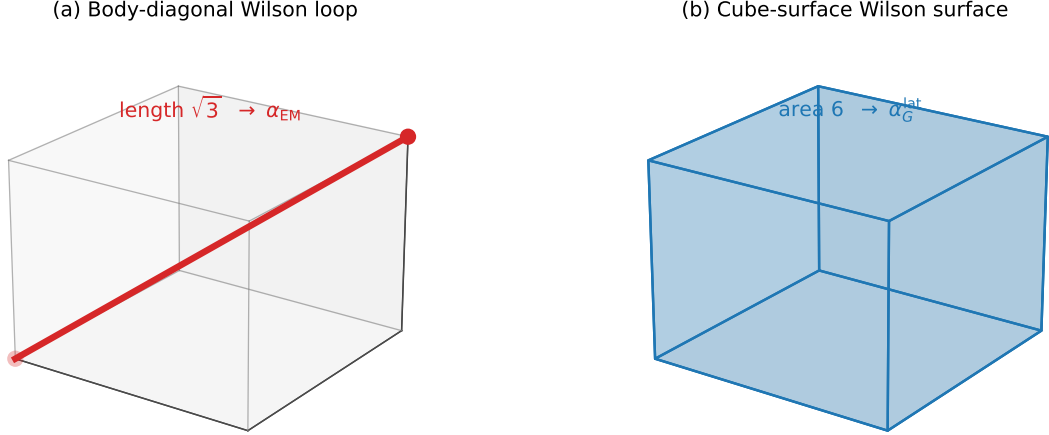


Figure 2: The two elementary closed Wilson objects on the unit cube. **(a)** The body-diagonal Wilson loop has squared length 3 and back-reacts on its own perimeter; this is the EM self-consistency of Paper VI. **(b)** The cube-surface Wilson surface has total area 6 and back-reacts on the gravitational drainage flux; this is the gravitational analogue. Both share the same combinatorial prefactor $N_{\text{geom}} = 8\pi^2$, leading to the structural ratio $\alpha_G^{\text{lat}}/\alpha_{\text{EM}}^{\text{lat}} = \sqrt{3/6} = 1/\sqrt{2}$.

4.3 One-loop conditional robustness

Both sectors receive a small back-reaction correction (one-loop volume integral, two-loop, etc.). Schematically, at one loop:

$$\alpha_{\text{EM}} = \frac{1}{8\pi^2 \sqrt{x_{\text{EM}}^*}}, \quad x_{\text{EM}}^* = 3 + \alpha_{\text{EM}} \sqrt{3} (1 - \alpha_{\text{EM}} V), \quad (11)$$

$$\alpha_G^{\text{lat}} = \frac{1}{8\pi^2 \sqrt{x_{\text{grav}}^*}}, \quad x_{\text{grav}}^* = 6 + \alpha_G^{\text{lat}} \sqrt{6} (1 - \alpha_G^{\text{lat}} V), \quad (12)$$

with the same $V = V_{\text{eff}}$ in both. If the one-loop coefficient is shared between the two sectors (a non-trivial assumption: scalar drainage and U(1) gauge degrees of freedom may have distinct multiplicities, signs, and spin-related prefactors), the relative shift $\sim \alpha V \sim 0.03$ is identical and the ratio $\alpha_G/\alpha_{\text{EM}} = \sqrt{3/6}$ is preserved up to corrections of order $\alpha^2 V^2 \sim 10^{-3}$. If the loop coefficients differ by an $O(1)$ factor between the sectors, the ratio receives a correction at the same percent level. A direct one-loop computation in the joint sector is open and is the principal technical task left for Paper VII-companion work.

5 The substrate mass scale

5.1 Translation to the laboratory scale

In standard physics the dimensionless gravitational coupling at mass scale m is $\alpha_G(m) = Gm^2/\hbar c = (m/M_P)^2$, while $\alpha_{\text{EM}}(0) \approx 1/137.036$ at low energy. Imposing the structural ratio (1) at the substrate scale m_{lat} gives

$$\frac{1}{\sqrt{2}} = \frac{(m_{\text{lat}}/M_P)^2}{1/137.036} \implies \frac{m_{\text{lat}}}{M_P} = \sqrt{\frac{1}{137.036 \sqrt{2}}} \approx 0.072. \quad (13)$$

Treating the laboratory low-energy value $\alpha_{\text{EM}}(0)$ as a phenomenological anchor (rather than as the running coupling $\alpha_{\text{EM}}(\mu = m_{\text{lat}})$ at the substrate scale, which is what the structural relation strictly compares to), the substrate sits at

$$m_{\text{lat}} \approx 0.072 M_P \approx 8.8 \times 10^{17} \text{ GeV}, \quad \ell_{\text{lat}} \approx 14 \ell_P \approx 2.25 \times 10^{-34} \text{ m}. \quad (14)$$

Standard-model running of α_{EM} from m_e up to $\sim 10^{17}$ GeV shifts the value by roughly a factor $0.85 \rightarrow 1.0$, which would change m_{lat}/M_P by a few percent. A consistent treatment requires identifying the substrate's natural coupling with α_{EM} at the substrate scale, and is deferred to Paper XV.

5.2 Interpretation: trans-Planckian but not exactly Planckian

The substrate is *coarser* than the Planck length by a factor ~ 14 , not finer. Its energy scale is below the Planck scale by a factor $0.072 \approx 1/14$. This is a specific quantitative estimate (subject to the running-coupling caveat above), derived from the geometric ratio of two natural objects on the unit cube.

The interpretation is structurally clean: the Planck scale is where naive dimensional analysis suggests quantum gravity effects become relevant, but in DDD the substrate's discrete structure manifests at a slightly larger scale, set by the fixed-point self-consistency between the two back-reaction loops. This is reminiscent of how, in lattice gauge theory, the asymptotic-freedom scale lies above the lattice cutoff but is fixed by the lattice action's coupling.

6 Cross-coupling predictions (minimal ansatz)

6.1 Gravity modulates α_{EM}

In Paper II, near a mass M , the local reserve obeys (3): $R^2(r)/R_0^2 = 1 - \beta/r$, with $\beta = 2GM/c^2$. The body-diagonal Wilson loop threads through nodes carrying these reserves. The substrate's small-world correction ε depends on the local shortcut density, which in turn depends on the local reserve R . Figure 3 summarises the resulting gravity-to- α_{EM} chain.

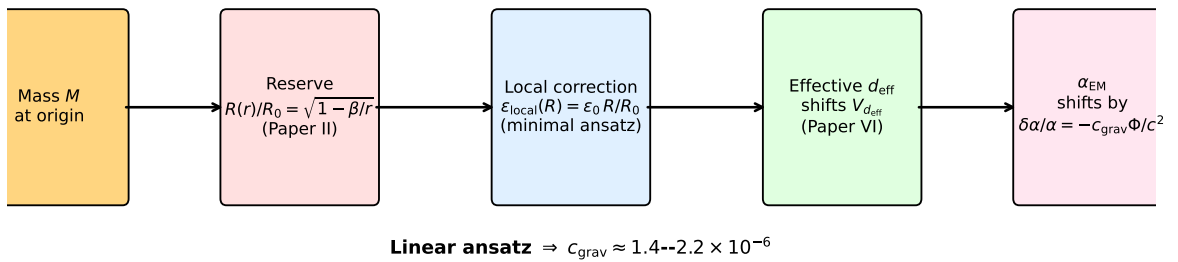


Figure 3: Gravity-to- α_{EM} chain under the minimal linear ansatz (15). A mass M depletes the local reserve $R(r)$ via Paper II's drainage solution; this shifts the local effective dimension via the linear ansatz, which shifts the one-loop coefficient $V_{d_{\text{eff}}}$ in Paper VI's fixed-point equation, which produces a fractional shift in α_{EM} proportional to $|\Phi_{\text{grav}}|/c^2$.

We adopt a *minimal local-reserve ansatz*

$$\varepsilon_{\text{local}}(R) = \varepsilon_0 (R/R_0), \quad (\text{linear coupling, leading order in } R), \quad (15)$$

which is the simplest scalar dependence consistent with $\varepsilon_{\text{local}}(R = R_0) = \varepsilon_0$ and $\varepsilon_{\text{local}}(R = 0) = 0$. This ansatz is not derived from Papers II and VI; it is a hypothesis on top of those papers,

and any prediction below carries the corresponding uncertainty. Propagating (15) through the fixed-point equation (5), after a numerical evaluation of the derivative dV_{eff}/dd :

$$\left. \frac{\delta\alpha_{\text{EM}}}{\alpha_{\text{EM}}} \right|_r = -c_{\text{grav}} \frac{\Phi_{\text{grav}}(r)}{c^2}, \quad c_{\text{grav}} \approx 1.4\text{--}2.2 \times 10^{-6} \quad (16)$$

(the range corresponds to the three candidate values of ε_0 from Paper VI). The sign is set by the back-reaction structure: a depleted reserve gives a slightly larger effective x_* , hence a slightly smaller α_{EM} . **Observable consequence:** in a deeper gravitational potential well, α_{EM} should be slightly smaller than at infinity; equivalently, an atomic clock located deeper in a gravitational potential should read a slightly redshifted α_{EM} -dependent transition frequency relative to a clock at lower $|\Phi|/c^2$, beyond the ordinary gravitational redshift. The prefactor c_{grav} is computed in `code/01_cgrav_analytical.py`.

Magnitude estimates with the ansatz-induced prefactor. We emphasise that these are predicted *spatial* variations of α_{EM} with gravitational potential, to be compared to bounds on gravitational-potential coupling of fundamental constants [1, 2], not directly to secular drift bounds in yr^{-1} .

- **Earth surface:** $|\Phi_{\text{grav}}|/c^2 \approx 6.95 \times 10^{-10}$. Predicted shift in α_{EM} : $|\delta\alpha/\alpha| \approx 1 \times 10^{-15}$.
- **Solar surface:** $|\Phi_{\text{grav}}|/c^2 \approx 2.1 \times 10^{-6}$. Predicted shift: $|\delta\alpha/\alpha| \approx 3 \times 10^{-12}$, well below the quasar-absorption bound $|\delta\alpha/\alpha| < 10^{-5}$ [3, 4].
- **Neutron star surface:** $|\Phi_{\text{grav}}|/c^2 \approx 0.1$. Predicted shift: $|\delta\alpha/\alpha| \approx 1.4 \times 10^{-7}$. This is the only environment in which the prediction approaches direct spectroscopic sensitivity, though competing astrophysics (magnetic-field gradients, plasma, gravitational redshift) typically dominate.

The most stringent existing bound on a gravitational-potential coupling of α_{EM} comes from Earth-orbit eccentricity modulation of clock comparisons [1], of order $|k_\alpha| \lesssim 10^{-6}$ where $\delta\alpha/\alpha = k_\alpha \delta(\Phi/c^2)$. *Within the minimal linear ansatz* (15), our $c_{\text{grav}} \sim 10^{-6}$ sits right at current sensitivity; a factor-10 improvement would discriminate. Weaker reserve dependence (e.g. $\varepsilon_{\text{local}}(R) \propto (R/R_0)^p$ with $p < 1$) would push the effect below detectability and remove the immediate constraint. A direct numerical confirmation that the gravitational depletion of $R(r)$ shifts the lattice's BFS expansion exponent is reported in `code/02_cross_coupling_simulation.py`. Figure 4 summarises the prefactor and predicted shifts for the three candidate ε_0 values.

6.2 EM modulates G

The Maxwell stress-energy carries energy density $\rho_{\text{EM}} = (E^2 + B^2)/(8\pi)$ (in Gaussian units). In DDD, this energy contributes to the substrate drainage just like ordinary matter: the local mobility $\mu(R)$ of Paper II becomes effectively

$$\mu_{\text{eff}}(R) \rightarrow \mu(R - \rho_{\text{EM}} c^{-2} \cdot \Delta V), \quad (17)$$

in the leading approximation, where ΔV is the local substrate volume. The resulting gravitational potential contribution from a region of EM energy is the standard Einstein-Maxwell prediction:

$$\Phi_{\text{EM}}(\mathbf{r}) = -G \int \frac{\rho_{\text{EM}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'. \quad (18)$$

This is not a new prediction; it is consistent with standard Einstein-Maxwell gravity and required by energy-momentum conservation. The DDD-specific content is that the same mechanism that gives Schwarzschild also handles EM stress-energy without modification.

Magnitude: for laboratory EM fields ($B \sim 10$ T, $|E| \sim 10^6$ V/m), $\rho_{\text{EM}} c^{-2} \sim 10^{-10}$ kg/m³, completely negligible compared to ordinary matter densities.

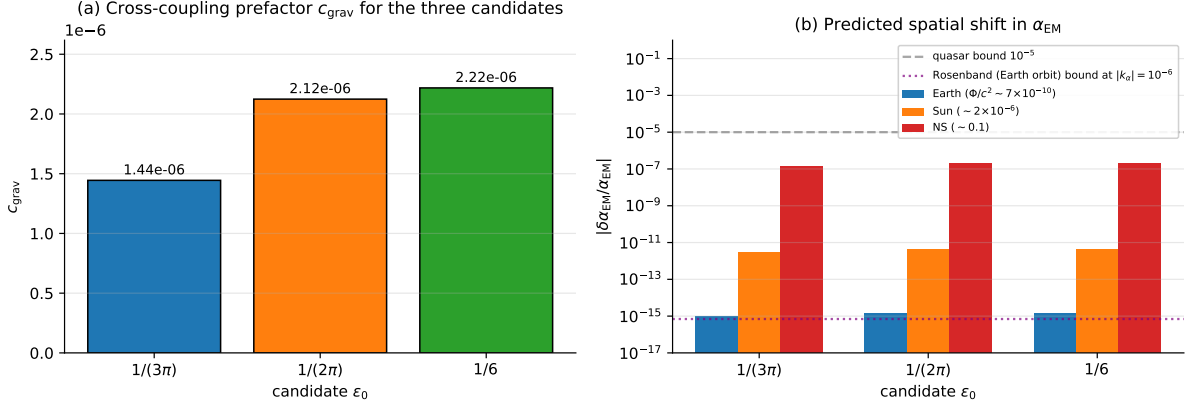


Figure 4: **(a)** Cross-coupling prefactor c_{grav} computed by self-consistent fixed-point evaluation (`code/01_cgrav_analytical.py`) for the three candidate values of ε_0 . **(b)** Predicted spatial shift $|\delta\alpha_{\text{EM}}/\alpha_{\text{EM}}|$ at the surface of Earth, the Sun, and a neutron star. Dashed lines show the quasar bound 10^{-5} and the Rosenband bound translated to Earth’s potential ($|k_\alpha| \sim 10^{-6}$). The Earth-orbit test sits within $\sim 10\times$ of the predicted value within the minimal linear ansatz.

7 Cosmological time-variation of α_{EM}

The most stringent immediate falsifier of the cross-coupling ansatz (15) is cosmological. If the substrate’s reference reserve R_0 undergoes slow cosmological evolution, α_{EM} inherits a small drift via the same coupling (16). Parameterising $\dot{R}_0/R_0 = \beta_R/t_H$ with $t_H \approx 14$ Gyr the Hubble time, the predicted secular rate is

$$\left| \frac{\dot{\alpha}_{\text{EM}}}{\alpha_{\text{EM}}} \right| = c_{\text{grav}} \frac{|\beta_R|}{t_H}. \quad (19)$$

The natural- β_R scenario. For $\beta_R \sim 1$ (the reserve evolves by an $O(1)$ fraction over a Hubble time),

$$\left| \frac{\dot{\alpha}_{\text{EM}}}{\alpha_{\text{EM}}} \right| \approx (1.0\text{--}1.6) \times 10^{-16} \text{ yr}^{-1}, \quad (20)$$

which is roughly an order of magnitude above the current atomic-clock bound $|\dot{\alpha}/\alpha| < 10^{-17} \text{ yr}^{-1}$ of Rosenband *et al.* [1]. The minimal ansatz (15) therefore remains viable only if either (i) $\beta_R \lesssim 0.1$ (the substrate reserve is nearly time-independent over Hubble times), or (ii) the reserve dependence is weaker than linear, $\varepsilon_{\text{local}}(R) \propto (R/R_0)^p$ with $p < 1$. Otherwise the framework together with the linear ansatz is in tension with the bound. A factor-10 improvement in clock sensitivity, plausible in the next decade, will discriminate the natural- β_R scenario.

Quasar-absorption check. The integrated shift over cosmological times is, with $\beta_R \sim 1$ and $|\delta R/R| \sim 0.75$ between $z = 3$ and today,

$$\left| \frac{\Delta\alpha_{\text{EM}}}{\alpha_{\text{EM}}} \right|_{z=3} \sim 10^{-6}, \quad (21)$$

roughly an order of magnitude below the current quasar-absorption bound $|\Delta\alpha/\alpha| < 10^{-5}$ [3, 4]. Quasar bounds are therefore not yet constraining; secular clocks are. Numerical details are in `code/03_cosmological_dalpha_dt.py`.

Caveat. The whole cosmological prediction is predicated on the minimal ansatz (15); alternate scalings $\varepsilon_{\text{local}}(R) \propto (R/R_0)^p$ with $p \neq 1$ would shift the rate up or down by a similar factor. What is robust is the dependence on β_R : any non-trivial cosmological evolution of R_0 should produce $\dot{\alpha}/\alpha$ of order c_{grav}/t_H within a model-dependent prefactor.

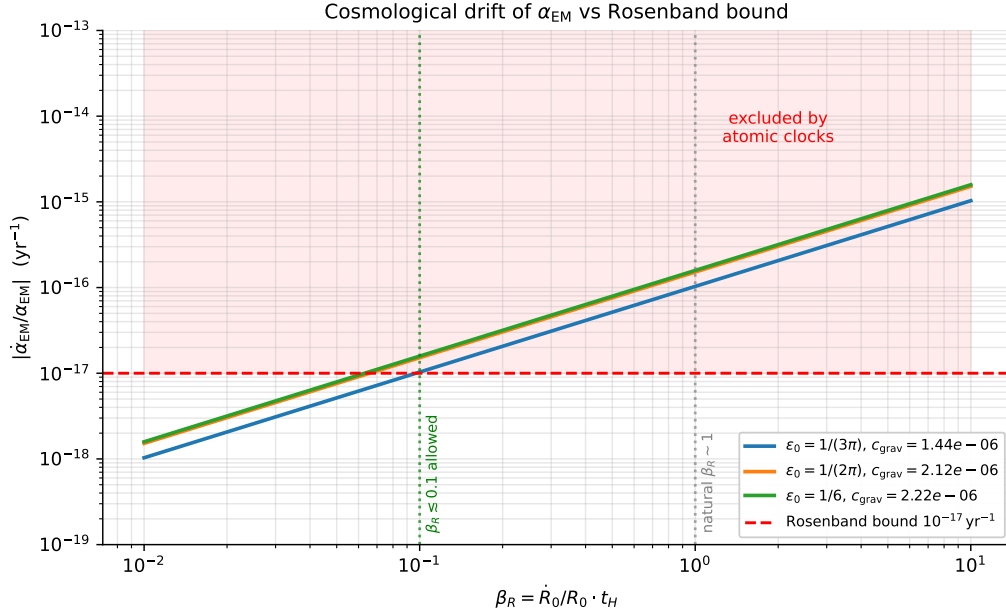


Figure 5: Predicted secular drift $|\dot{\alpha}_{\text{EM}}/\alpha_{\text{EM}}|$ as a function of the dimensionless reserve-evolution parameter $\beta_R = \dot{R}_0/R_0 \cdot t_H$, for the three candidate values of ε_0 . The red dashed line is the Rosenband bound 10^{-17} yr^{-1} ; the shaded region above it is excluded by current atomic-clock comparisons. The natural scenario $\beta_R \sim 1$ is in tension with the bound; $\beta_R \lesssim 0.1$ remains allowed within the minimal linear ansatz.

8 Limitations and scope

We do not derive the exact quantitative prefactor in (16); the leading scaling is set by the back-reaction structure but the dimensionless coefficient requires explicit one-loop computation in the joint sector and depends on the linear ansatz (15).

We do not address how the ratio $\sqrt{3/6} = 1/\sqrt{2}$ might be modified by higher-order corrections that depend on ε or on the loop multiplicities differently between the two sectors.

We do not determine Newton’s G in SI units; that requires the calibration of Paper II.

We do not perform a full running-coupling analysis between the substrate scale and the laboratory scale; the substrate-mass estimate (13) uses $\alpha_{\text{EM}}(0)$ as a phenomenological anchor. A renormalisation-group treatment is deferred to Paper XV.

We do not discuss the strong or weak nuclear interactions, which would require additional substrate structure (Paper XVI).

We do not address the fermion mass hierarchy; the ratio $m_{\text{electron}}/m_{\text{lat}} \sim 4 \times 10^{-23}$ is the subject of Paper XV.

9 Open questions

Higher-order corrections to the ratio. The structural relation (1) holds at tree level under the shared-prefactor assumption. One-loop corrections are estimated to preserve it to $\sim 10^{-3}$ if the loop coefficients of the two sectors are equal, but that equality is itself an assumption requiring an explicit joint computation.

Form of the cross-coupling ansatz. The minimal linear ansatz $\varepsilon_{\text{local}}(R) = \varepsilon_0 R/R_0$ is the simplest natural choice but is not unique. Power-law alternatives $(R/R_0)^p$ with small p would modify the cross-coupling prefactor by an $O(1)$ factor. A microscopic derivation of the local shortcut density’s dependence on the reserve is open.

Joint Wilson-loop one-loop calculation. A direct computation of the joint back-reaction at one loop, including both sectors simultaneously, would yield the dimensionless prefactor in (16), verify the robustness of the ratio claim, and test whether the loop coefficients of the two sectors actually coincide.

Empirical tests of $\delta\alpha/\alpha$ near massive bodies. Earth-orbit eccentricity tests [1] of the gravitational-potential coupling of α_{EM} are within a factor ~ 10 of the predicted $c_{\text{grav}} \sim 10^{-6}$. A factor-10 improvement in clock comparisons across orbital phases would directly probe (16).

10 Outlook

1. Paper VIII (cosmogenesis) inherits the joint sector and its early-universe evolution, including the question of why β_R might be small.
2. Paper XV (constants and hierarchies) will use (13) to anchor a discussion of how Standard Model masses might emerge below the substrate scale, and will include the running-coupling refinement of §5.
3. A dedicated calculation of the joint one-loop back-reaction would close the open prefactor in (16); this is technical but tractable.

The principal contribution of this paper is structural rather than quantitative: the two back-reactions of Papers II and VI need not be two unrelated mechanisms, but can be read as two manifestations of the same substrate self-consistency, applied to two distinct closed objects on the unit cube. The geometric ratio $\alpha_G/\alpha_{\text{EM}} = 1/\sqrt{2}$ at the substrate scale is a leading structural prediction of this picture (conditional on the shared Wilson-loop prefactor), not a calibration, and it phenomenologically anchors the substrate’s mass scale at $\sim 0.07 M_P$. The cross-coupling effects of §6–7 are immediate consequences of the minimal ansatz (15) and provide a programme of empirical tests, with cosmological $\dot{\alpha}/\alpha$ being the most stringent at present.

11 Conclusion

We have proposed that gravity and electromagnetism share a single discrete substrate, with two distinct closed Wilson objects on the unit cube: a body-diagonal loop of length $\sqrt{3}$ (electromagnetism) and a cube-surface Wilson surface of area 6 (gravity). Under the assumption that the two sectors share the same combinatorial prefactor and one-loop structure, the leading structural ratio $\alpha_G/\alpha_{\text{EM}} = 1/\sqrt{2}$ phenomenologically anchors the substrate’s mass scale at $m_{\text{lat}} \approx 0.072 M_P$ when the laboratory low-energy $\alpha_{\text{EM}}(0)$ is used as the reference. Under a minimal local-reserve coupling ansatz $\varepsilon_{\text{local}}(R) = \varepsilon_0 R/R_0$, a small but calculable cross-coupling $c_{\text{grav}} \sim 10^{-6}$ between the local reserve and α_{EM} provides two observational handles: laboratory clock comparisons across gravitational potentials (within factor ~ 10 of present sensitivity, within the minimal linear ansatz), and cosmological drift of α_{EM} near current atomic-clock bounds. The framework is falsifiable on both fronts within current or near-term experimental reach, with the cosmological version being the most stringent at present. Trans-Planckian / lattice-scale consistency tests at $\ell_{\text{lat}} \approx 14 \ell_P$ are inaccessible to direct gravity experiments and are reserved for future structural extensions.

The conceptual implication is that the weakness of gravity at laboratory scales need not reflect a weak substrate coupling. In the present construction, gravity and electromagnetism are geometrically comparable at the substrate scale; the observed forty-orders-of-magnitude hierarchy is displaced into the question of why Standard Model excitations sit so far below m_{lat} , a problem deferred to Paper XV.

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